

HR Analytics Report

1. Data Loading

- Imported the Pandas library for data manipulation.
- Loaded the dataset **HR-Employee-Attrition.csv** into a DataFrame.
- Verified successful loading of employee records.

2. Initial Data Inspection

Performed an initial assessment of the dataset using:

- `df.shape` – Checked the number of rows and columns.
- `df.info()` – Examined data types and non-null values.
- `df.head()` – Viewed sample employee records.
- `df.describe(include='all')` – Generated summary statistics for numerical and categorical features.

3. Data Quality Checks

Missing Value Analysis

- Checked all columns for missing values using:
 - `df.isnull().sum()`
- Confirmed data completeness before analysis.

Duplicate Record Analysis

- Identified duplicate employee records using:
 - `df.duplicated().sum()`
- Ensured data consistency and uniqueness.

4. Data Cleaning

Removing Constant Columns

The following columns contained only a single value across all records and did not contribute to analysis:

- EmployeeCount
- Over18
- StandardHours

These columns were removed to improve data quality and reduce redundancy.

5. Data Transformation

Standardizing Column Names

- Converted all column names to lowercase format.
- Improved consistency for SQL queries and Power BI integration.

Encoding Attrition

Converted the Attrition column into numeric format:

Original Value	Encoded Value
Yes	1
No	0

This enabled quantitative analysis and correlation calculations.

Encoding Overtime

Converted Overtime into numeric format:

Original Value	Encoded Value
Yes	1
No	0

6. Feature Engineering

Age Group Creation

Created a new feature called **AgeGroup** using age ranges:

- 18–25
- 26–35
- 36–45
- 46–55
- 56–65

Purpose:

- Employee demographic segmentation.
- Attrition analysis by age category.

□

Salary Band Creation

Created a new feature called **SalaryBand** based on Monthly Income:

- Low
- Medium
- High
- Very High

Purpose:

- Salary-based workforce segmentation.
- Attrition analysis by compensation level.

7. Exploratory Data Analysis (EDA)

Outlier Detection

- Used Box Plot visualization on Monthly Income.
- Identified potential salary outliers.

Numerical Data Summary

Analyzed key numerical variables:

- Age
- Monthly Income
- Years at Company
- Distance From Home

Generated descriptive statistics to understand employee characteristics.

8. Employee Attrition Analysis

Attrition Rate Calculation

Calculated overall attrition percentage across the organization.

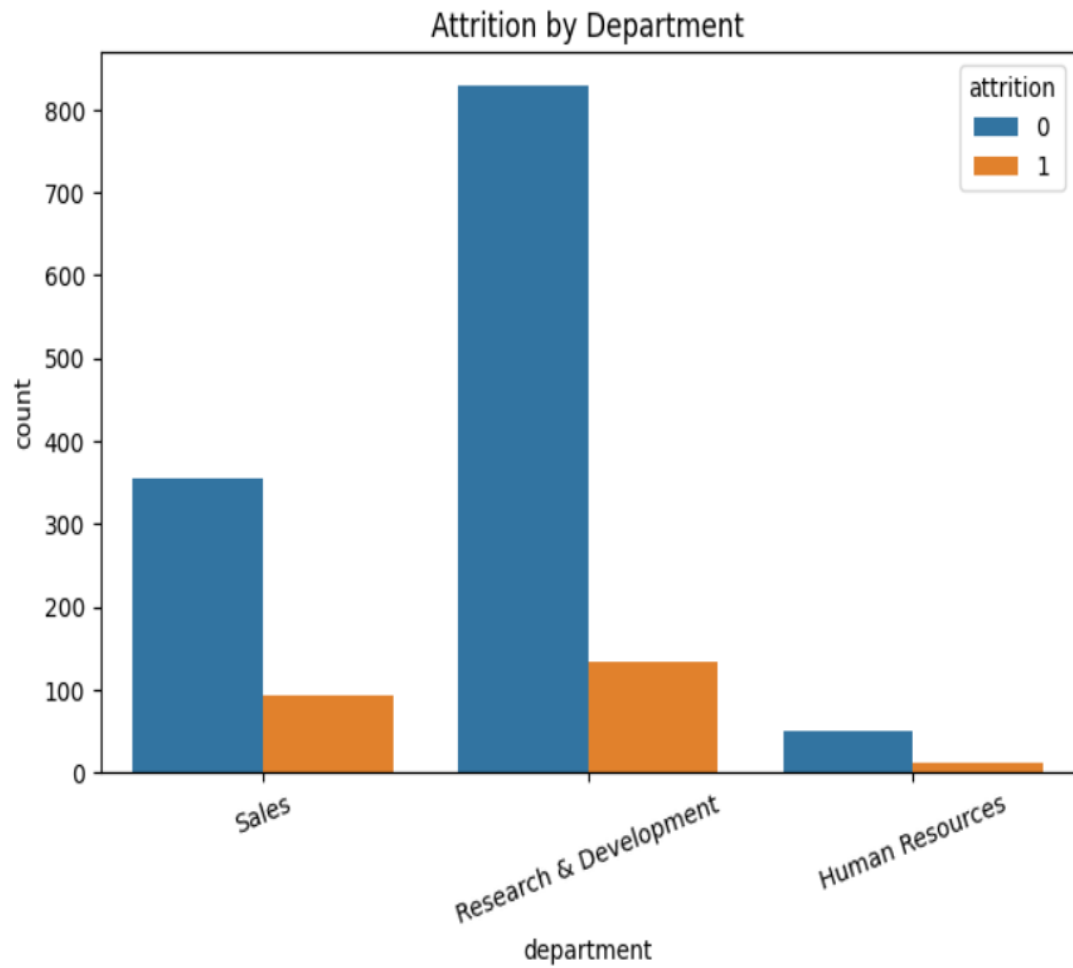
Visualization:

- Pie Chart showing:
 - Employees who left
 - Employees who stayed

Attrition by Department

Analyzed attrition patterns across:

- Sales



- Research & Development
- Human Resources

Visualization:

- Department-wise Attrition Count Plot.

Attrition by Gender

Compared attrition rates between:

- Male Employees
- Female Employees

Visualization:

- Gender vs Attrition Count Plot.

Overtime vs Attrition

Investigated the relationship between:

- Overtime Status
- Employee Attrition

Visualization:

- Overtime vs Attrition Count Plot.

Salary vs Attrition

Examined whether compensation impacts employee turnover.

Visualization:

- Monthly Income Box Plot by Attrition Status.

9. Workforce Demographic Analysis

Gender Distribution

Analyzed employee distribution by gender.

□

Visualization:

Gender Count Plot.

Department-wise Employee Distribution

Analyzed workforce concentration across departments.

Visualization:

□ Department Count Plot.

Job Role Distribution

Studied employee counts across job roles such as:

- Sales Executive
- Research Scientist
- Laboratory Technician
- Manufacturing Director
- Healthcare Representative Visualization:
- Job Role Distribution Chart.

Age Distribution

Analyzed employee age patterns.

Visualization:

□ Histogram with Density Curve.

Salary Distribution

Analyzed Monthly Income distribution.

Visualization:

□ Histogram with KDE Curve.

10. Correlation Analysis

Generated a Correlation Heatmap for all numerical variables.

□

Purpose:

Identify relationships among employee factors.

- Determine variables influencing attrition.
- Support predictive modeling and business insights.

11. MySQL Database Integration

Database Connection

Connected Python with MySQL using:

- PyMySQL
- SQLAlchemy Database Name:
- hr_analytics

Data Storage

Loaded the cleaned dataset into MySQL table:

- employees

Benefits:

- Centralized storage.
- SQL-based analysis.
- Power BI connectivity.

12. Data Analysis Using SQL

Database Setup

Database Creation

```
CREATE DATABASE hr_analytics;  
USE hr_analytics;
```

Purpose:

- Create a dedicated database for HR analytics.
- Store employee records for structured querying and reporting.

Employee Workforce Analysis

Total Employees

```
SELECT COUNT(*) FROM employees;
```

Result Grid		Filter Rows:
	COUNT(*)	
▶	1470	

Overall Attrition Rate

Result Grid		Filter Rows:
	Attrition_Rate	
▶	83.88	

Department-wise Attrition

	Department	Attrition_Count
▶	Research & Development	828
	Sales	354
	Human Resources	51

Attrition by Gender

	Gender	Attrition_Count
▶	Male	732
	Female	501

□

Attrition Rate by Job Role

	JobRole	AttritionRate
▶	Sales Executive	82.52
	Research Scientist	83.90
	Laboratory Technician	76.06
	Manufacturing Director	93.10
	Healthcare Representative	93.13
	Manager	95.10
	Sales Representative	60.24
	Research Director	97.50
	Human Resources	76.92

Average Salary by Department

	Department	ROUND(AVG(MonthlyIncome),2)
▶	Sales	6959.17
	Research & Development	6281.25
	Human Resources	6654.51

Top 5 Highest Paid Employees

	EmployeeNumber	MonthlyIncome
▶	259	19999
	1035	19973
	1191	19943
	226	19926
	787	19859

Salary Band Analysis

	SalaryBand	EmployeeCount
▶	Medium	440
	Low	749
	High	281

□

Average Years at Company by Department

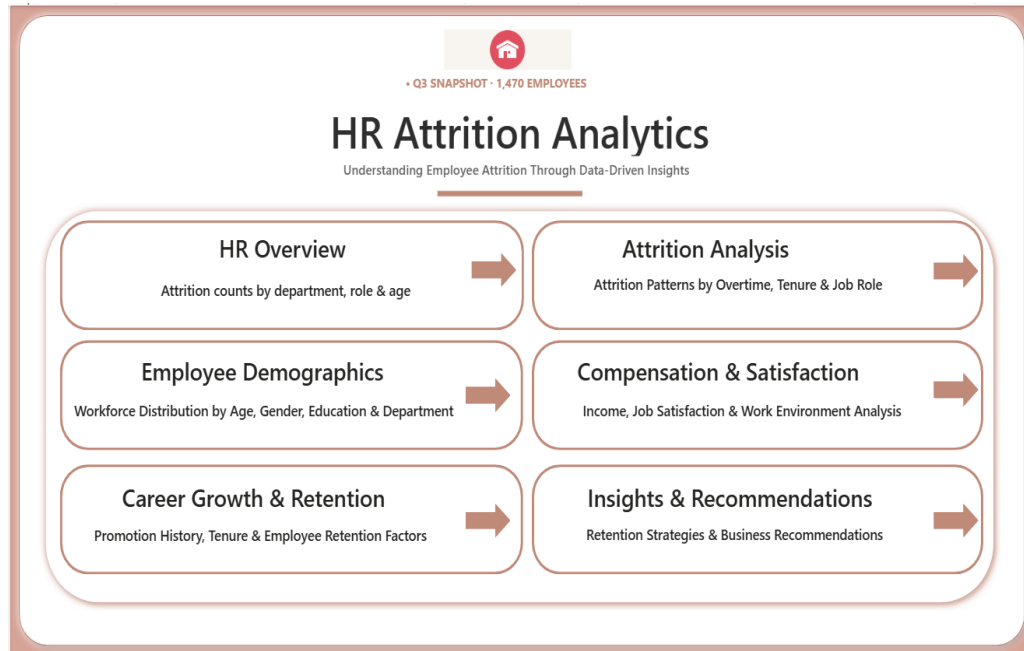
	Department	AVG(YearsAtCompany)
▶	Sales	7.2848
	Research & Development	6.8647
	Human Resources	7.2381

Impact of Overtime on Attrition

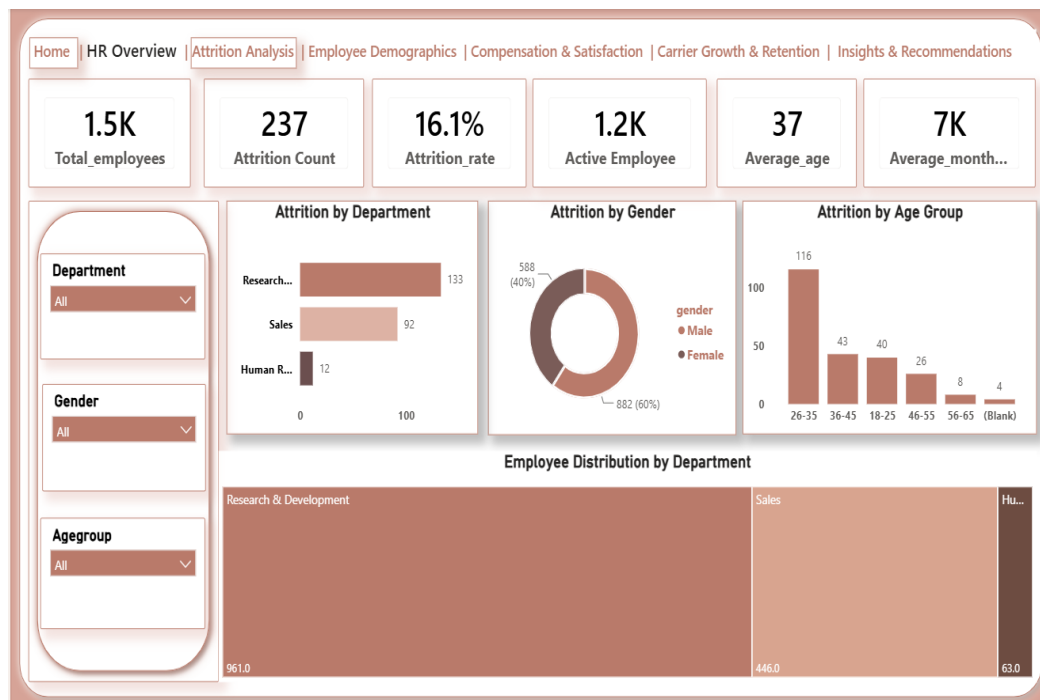
Result Grid   Filter Rows:			
	OverTime	Employees	Attrition
▶	1	416	289
	0	1054	944

13. Dashboard in Power BI

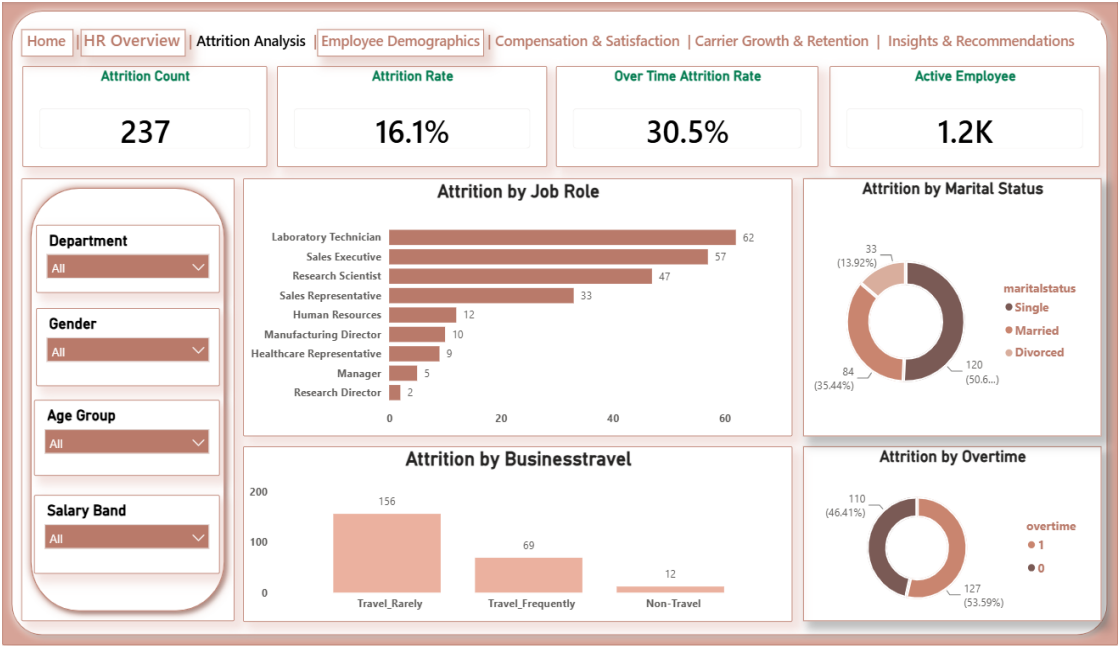
Home Page



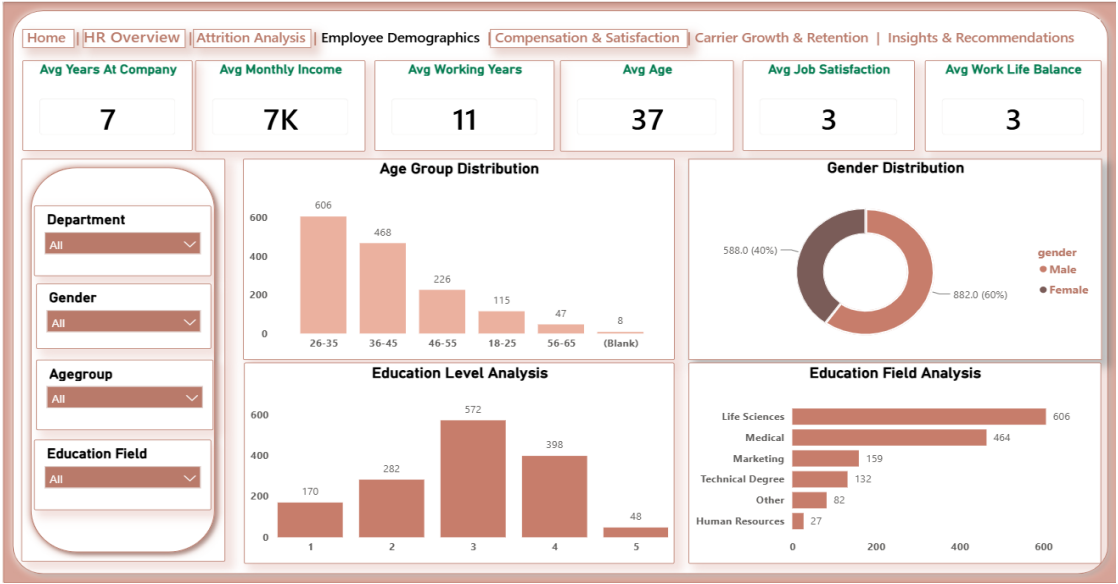
HR Overview Page



Attrition Analysis Page



Employee Demography Page



Compensation & Satisfaction Page



Career Growth & Retention Page





Insights & Recommendations Page

Home

HR Overview

Attrition Analysis

Employee Demographics

Compensation & Satisfaction

Carrier Growth & Retention

Insights & Recommendations

Attrition Rate

16.1%

Attrition Count

237

Overtime Attrition Rate

30.5%

Highest Risk Department

Sales

Highest Risk Job Role

Sales Representative

Key Insights

- The overall attrition rate is **16.1%**, with **237 employees** leaving the company.
- Employees working overtime have a significantly higher **attrition rate (30.5%)** compared to employees who do not work overtime (**10.4%**).
- Sales Department** has the highest attrition rate (**20.6%**) among all departments.
- Sales Representatives** experience the highest attrition rate (**39.8%**), making them the most vulnerable employee group.
- Employees in the **Low Salary Band** show the highest attrition rate (**21.8%**), indicating compensation is a major retention factor.
- Employees with **Job Satisfaction Level 1** have an attrition rate of **22.8%**, almost double that of highly satisfied employees.
- Attrition tends to be higher among employees experiencing longer periods without promotion, highlighting the importance of career progression opportunities.

Key Recommendations

- Improve workload distribution, staffing levels, and workforce planning to reduce employee burnout.
- Launch targeted retention initiatives for the Sales department, particularly for Sales Representatives.
- Reassess salary structures and benefits for employees in lower salary bands to improve engagement and retention.
- Strengthen employee recognition programs, feedback mechanisms, and workplace culture initiatives.
- Create clear career paths, mentorship programs, and internal mobility opportunities.
- Improve onboarding, training, and first-year support programs to reduce early-stage turnover.
- Use HR analytics to proactively identify employees with overtime exposure, low salary levels, low satisfaction scores, and limited career growth.
- Implement predictive attrition monitoring dashboards to identify at-risk employees before resignation occurs.